

Problem

Rework Prob. 3.19 using mesh analysis.

Step-by-step solution

Step 1 of 8

Consider the following circuit diagram:

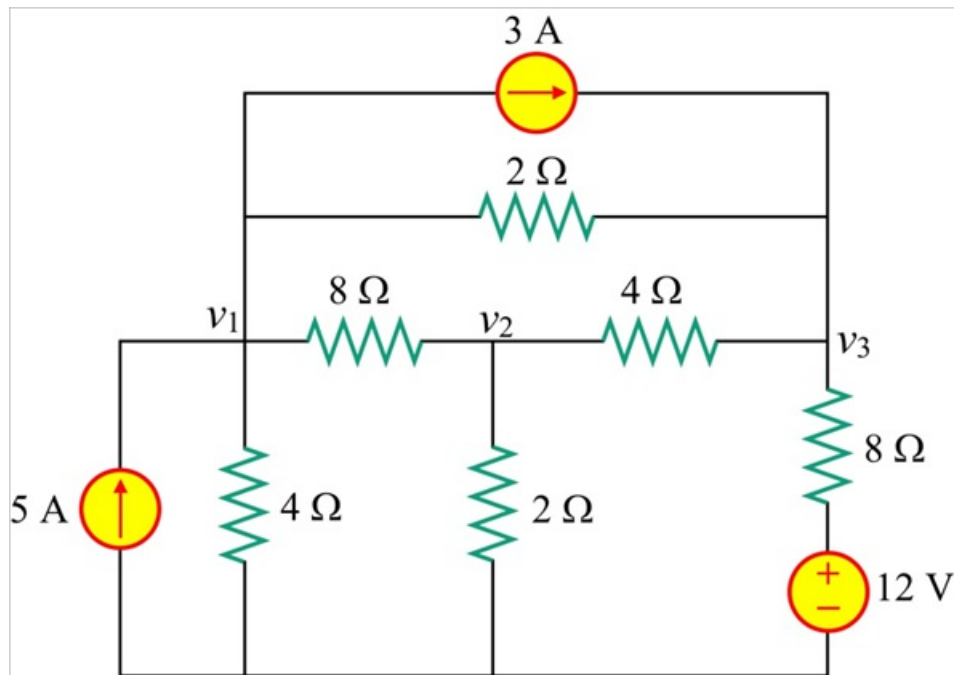


Figure 1

Step 2 of 8

Draw the circuit diagram with mesh notations:

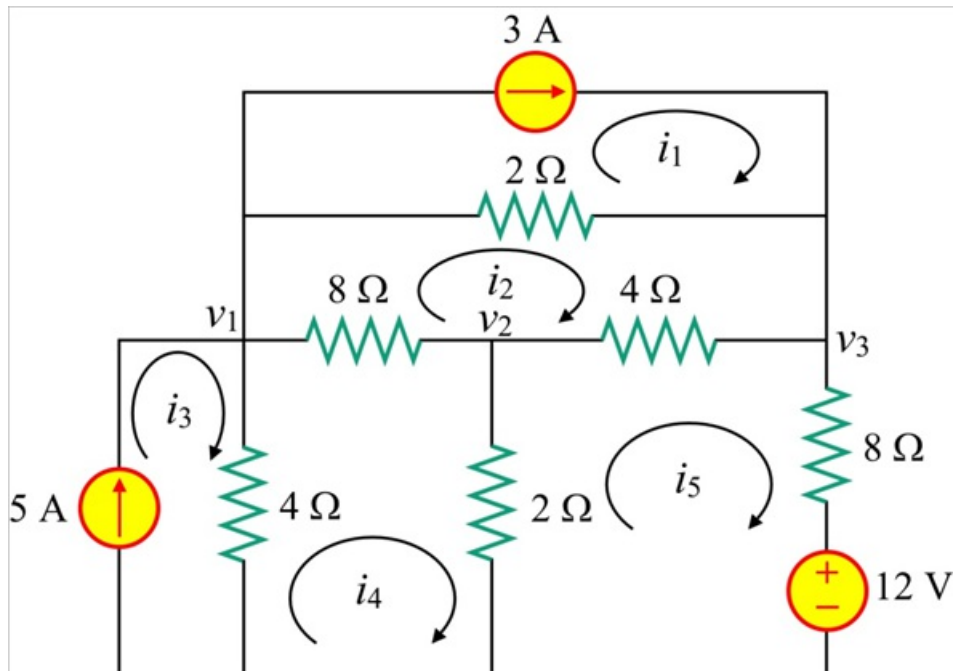


Figure 2

Step 3 of 8

From the figure 1, the current in the mesh 1 is 3 A and current in the mesh is 5 A.

$$i_1 = 3 \text{ A} \dots\dots (1)$$

$$i_3 = 5 \text{ A} \dots\dots (2)$$

Apply Kirchhoff's voltage law to the mesh 2.

$$2(i_2 - i_1) + 8(i_2 - i_4) + 4(i_2 - i_5) = 0$$

$$-2i_1 + (2 + 8 + 4)i_2 - 8i_4 - 4i_5 = 0$$

$$-2i_1 + 14i_2 - 8i_4 - 4i_5 = 0$$

$$i_1 - 7i_2 + 4i_4 + 2i_5 = 0 \dots\dots (3)$$

Step 4 of 8

Apply Kirchhoff's voltage law to the mesh 4.

$$4(i_4 - i_3) + 8(i_4 - i_2) + 2(i_4 - i_5) = 0$$

$$-8i_2 - 4i_3 + (4 + 8 + 2)i_4 - 2i_5 = 0$$

$$-8i_2 - 4i_3 + 14i_4 - 2i_5 = 0$$

$$4i_2 + 2i_3 - 7i_4 + i_5 = 0 \dots\dots (4)$$

Apply Kirchhoff's voltage law to the mesh 5.

$$2(i_5 - i_4) + 4(i_5 - i_2) + 8i_5 + 12 = 0$$

$$-4i_2 - 2i_4 + (2 + 4 + 8)i_5 = -12$$

$$-4i_2 - 2i_4 + 14i_5 = -12$$

$$2i_2 + i_4 - 7i_5 = 6 \dots\dots (5)$$

Step 5 of 8

Write equations (1), (2), (3), (4), and (5) in the matrix form:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & -7 & 0 & 4 & 2 \\ 0 & 4 & 2 & -7 & 1 \\ 0 & 2 & 0 & 1 & -7 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 0 \\ 0 \\ 6 \end{bmatrix}$$

Step 6 of 8

The following is the MATLAB code for calculating loop currents.

```
>> a=[1 0 0 0 0;0 0 1 0 0;1 -7 0 4 2;0 4 2 -7 1;0 2 0 1 -7];
```

```
b=[3;5;0;0;6];
```

```
c=inv(a)*b
```

MATLAB output:

```
c =
```

```
3.0000
```

```
1.8667
```

```
5.0000
```

```
2.5000
```

```
0.0333
```

Step 7 of 8

The values of the mesh currents are,

$$i_1 = 3 \text{ A}$$

$$i_2 = 1.8667 \text{ A}$$

$$i_3 = 5 \text{ A}$$

$$i_4 = 2.5 \text{ A}$$

$$i_5 = 33.3 \text{ mA}$$

Calculate the node voltage v_1 .

$$v_1 = 4(i_3 - i_4)$$

$$= 4(5 - 2.5)$$

$$= 4(2.5)$$

$$= 10 \text{ V}$$

Step 8 of 8

Calculate the node voltage v_2 .

$$\begin{aligned}v_2 &= 2(i_4 - i_5) \\&= 2(2.5 - 0.033) \\&= 2(2.4667) \\&= 4.9334 \text{ V}\end{aligned}$$

Calculate the node voltage v_3 .

$$\begin{aligned}v_3 &= 8i_5 + 12 \\&= (8)(0.0333) + 12 \\&= 0.267 + 12 \\&= 12.267 \text{ V}\end{aligned}$$

Therefore, the node voltages, v_1, v_2, v_3 are 10 V, 4.9334 V, 12.267 V.